

DBSE WEEKLY LAB EXPERIMENTS

KAIRAV VASHI

2520030582

SECTION – 7

CSE

JOIN AND SET OPERATIONS

The screenshot shows the SQL Studio interface. The SQL Editor contains the following script:

```
1 CREATE DATABASE IF NOT EXISTS practice_db;
2 USE practice_db;
3
4 DROP TABLE IF EXISTS first_table;
5 DROP TABLE IF EXISTS second_table;
6 DROP TABLE IF EXISTS class_info;
7 DROP TABLE IF EXISTS class;
8
9 CREATE TABLE class (
10     id INT,
11     name VARCHAR(30))
```

The Result Grid shows the following data:

id	name	id	address
1	Kairav	1	DELHI
2	Kairav	2	MUMBAI
3	Kairav	3	CHENNAI
7		7	NOIDA
8		8	PANIPAT

The Action Output shows the following messages:

#	Time	Action	Message	Duration / Fetch
1	09:28:45	CREATE DATABASE IF NOT EXISTS practice_db	1 row(s) affected	0.015 sec
2	09:28:45	USE practice_db	0 row(s) affected	0.000 sec

The screenshot shows the SQL Studio interface. The SQL Editor contains the following script:

```
9 CREATE TABLE class (
10     id INT,
11     name VARCHAR(30))
12 );
13
14 CREATE TABLE class_info (
15     id INT,
16     address VARCHAR(30))
17 );
18
19 INSERT INTO class VALUES (1,'Kairav'), (2,'Kairav'), (3,'Kairav'), (4,'Kairav'), (5,'Kairav')
```

The Result Grid shows the following data:

id	name	id	address
1	Kairav	1	DELHI
2	Kairav	2	MUMBAI
3	Kairav	3	CHENNAI
7		7	NOIDA
8		8	PANIPAT

The Action Output shows the following messages:

#	Time	Action	Message	Duration / Fetch
7	09:28:45	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.015 sec
8	09:28:45	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected	0.015 sec
9	09:28:45	INSERT INTO class VALUES (1,'Kairav'), (2,'Kairav'), (3,'Kairav'), (4,'Kairav'), (5,'Kairav')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec

Navigator: exp2 SQL File 3'

SCHEMAS
Filter objects
sys

Administration Schemas
Information
No object selected

```

19 • INSERT INTO class VALUES
20 (1,'Kairav'),
21 (2,'Kairav'),
22 (3,'Kairav'),
23 (4,'Kairav'),
24 (5,'Kairav');
25
26 INSERT INTO class_info VALUES
27 (1,'DELHI'),
28 (2,'MUMBAI'),
29 (3,'CHENNAI'),

```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content: |

id	name	id	address
1	Kairav	1	DELHI
2	Kairav	2	MUMBAI
3	Kairav	3	CHENNAI
7		7	NOIDA
8		8	PANIPAT

Result 36 x Result 49

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	09:28:45	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.015 sec
8	09:28:45	CREATE TABLE class_info (id INT, address VARCHAR(30))	0 row(s) affected	0.015 sec
9	09:28:45	INSERT INTO class VALUES (1,'Kairav'), (2,'Kairav'), (3,'Kairav'), (4,'Kairav'), (5,'Kairav')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
10	09:28:45	INSERT INTO class_info VALUES (1,'DELHI'), (2,'MUMBAI'), (3,'CHENNAI'), (7,'NOIDA'), (8,'PANIPAT')	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

Navigator: exp2 SQL File 3'

SCHEMAS
Filter objects
sys

Administration Schemas
Information
No object selected

```

30 (7,'NOIDA'),
31 (8,'PANIPAT');
32
33 • SELECT *
34 FROM class
35 CROSS JOIN class_info;
36
37 • SELECT *
38 FROM class
39 INNER JOIN class_info
40 ON class.id = class_info.id;

```

Result Grid | Filter Rows: | Exports: | Wrap Cell Content: |

id	name	id	address
1	Kairav	1	DELHI
2	Kairav	2	MUMBAI
3	Kairav	3	CHENNAI
7		7	NOIDA
8		8	PANIPAT

Result 36 x Result 49

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	09:28:45	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	25 row(s) returned	0.000 sec / 0.000 sec
12	09:28:45	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
13	09:28:45	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Navigator

SCHEMAS

Filter objects

sys

Administration Schemas

Information

No object selected

SQL File 3'

Limit to 1000 rows

```

91 CREATE TABLE first_table (
92   id INT,
93   name VARCHAR(30)
94 );
95
96 CREATE TABLE second_table (
97   id INT,
98   name VARCHAR(30)
99 );
100
101 INSERT INTO first_table VALUES

```

Result Grid

id	name	id	address
4	Kairav	NOIDA	
5	Kairav	NOIDA	
7	NOIDA		
8	PANIPAT		

Output

#	Time	Action	Message	Duration / Fetch
16	09:32:56	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 1000	2 row(s) returned	0.000 sec / 0.000 sec
17	09:32:56	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id = class_info.id LIMIT 0.1000	5 row(s) returned	0.000 sec / 0.000 sec
18	09:32:56	SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL LIMIT 0.1000	2 row(s) returned	0.000 sec / 0.000 sec
19	09:32:56	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id UNION SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id	7 row(s) returned	0.000 sec / 0.000 sec
20	09:32:56	SELECT * FROM class LEFT JOIN class_info ON class.id = class_info.id WHERE class_info.id IS NULL UNION SELECT * FROM class RIGHT JOIN class_info ON class.id = class_info.id	4 row(s) returned	0.000 sec / 0.000 sec
21	09:32:56	CREATE TABLE first_table (id INT, name VARCHAR(30))	0 row(s) affected	0.016 sec
22	09:32:56	CREATE TABLE second_table (id INT, name VARCHAR(30))	0 row(s) affected	0.031 sec

Navigator

SCHEMAS

Filter objects

sys

Administration Schemas

Information

No object selected

SQL File 3'

Limit to 1000 rows

```

101 INSERT INTO first_table VALUES
102   (1,'Kairav'),
103   (2,'Kairav');
104
105 INSERT INTO second_table VALUES
106   (2,'Kairav'),
107   (3,'Kairav');
108
109 SELECT * FROM first_table
110 UNION
111 SELECT * FROM second_table;

```

Result Grid

id	name	id	address
4	Kairav	NOIDA	
5	Kairav	NOIDA	
7	NOIDA		
8	PANIPAT		

Output

#	Time	Action	Message	Duration / Fetch
24	09:32:56	INSERT INTO second_table VALUES (2,'Kairav'), (3,'Kairav')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
25	09:32:56	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
26	09:32:56	SELECT name FROM first_table UNION SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
27	09:32:56	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec

Navigator: exp2 SQL File 3*

SCHMAS

Filter objects

sys

Administration Schemas

Information

No object selected

```

113 SELECT name FROM first_table
114 UNION
115 SELECT name FROM second_table;
116
117 SELECT * FROM first_table
118 UNION ALL
119 SELECT * FROM second_table;
120
121 SELECT COUNT(*)
122 FROM (
123     SELECT * FROM first_table

```

Result Grid

id	name
1	Karav
2	Karav
3	Karav

Output

#	Time	Action	Message	Duration / Fetch
24	09:32:56	INSERT INTO second_table VALUES (2,'Karav'), (3,'Karav')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
25	09:32:56	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
26	09:32:56	SELECT name FROM first_table UNION SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
27	09:32:56	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
28	09:32:56	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table)	1 row(s) returned	0.000 sec / 0.000 sec
29	09:32:56	SELECT first_table.* FROM first_table INNER JOIN second_table ON first_table.id = second_table.id AND first...	1 row(s) returned	0.000 sec / 0.000 sec
30	09:32:56	SELECT first_table.name FROM first_table INNER JOIN second_table ON first_table.name = second_table.n...	4 row(s) returned	0.000 sec / 0.000 sec
31	09:32:56	SELECT first_table.* FROM first_table LEFT JOIN second_table ON first_table.id = second_table.id AND first...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Navigator: exp2 SQL File 3*

SCHMAS

Filter objects

sys

Administration Schemas

Information

No object selected

```

121 SELECT COUNT(*)
122 FROM (
123     SELECT * FROM first_table
124     UNION ALL
125     SELECT * FROM second_table
126 ) A;
127
128 SELECT first_table.*
129 FROM first_table
130 INNER JOIN second_table
131 ON first_table.id = second_table.id AND first_table.name = second_table.name;

```

Result Grid

id	name
1	Karav
2	Karav
3	Karav

Output

#	Time	Action	Message	Duration / Fetch
24	09:32:56	INSERT INTO second_table VALUES (2,'Karav'), (3,'Karav')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
25	09:32:56	SELECT * FROM first_table UNION SELECT * FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
26	09:32:56	SELECT name FROM first_table UNION SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
27	09:32:56	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
28	09:32:56	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table)	1 row(s) returned	0.000 sec / 0.000 sec
29	09:32:56	SELECT first_table.* FROM first_table INNER JOIN second_table ON first_table.id = second_table.id AND first...	1 row(s) returned	0.000 sec / 0.000 sec
30	09:32:56	SELECT first_table.name FROM first_table INNER JOIN second_table ON first_table.name = second_table.n...	4 row(s) returned	0.000 sec / 0.000 sec
31	09:32:56	SELECT first_table.* FROM first_table LEFT JOIN second_table ON first_table.id = second_table.id AND first...	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Navigation

SCHEMAS

Filter objects

sys

exp2

SQL File 2

Limit to 1000 rows

153

ON c.id = ci.id;

154

155

SELECT c.id,

156

c.name,

157

CASE

158

WHEN ci.address IS NULL THEN 'Address Missing'

159

ELSE 'Address Available'

160

END AS Status

161

FROM class c

162

LEFT JOIN class_info ci

163

ON c.id = ci.id;

Result Grid

Filter Rows:

Export:

Wrap Cell Contents: 15

Result 50

Result 51

Result 52

Result 53

Result 54

Result 55

Result 56

Result 57

Result 58

Result 59

Result 60

Result 61

Result 62

Result 63

Result 69

Read Only

Information

No object selected

Output

Action Output

#

Time

Action

Message

Duration / Fetch

28

09:32:56

SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table) ...

1 row(s) returned

0.000 sec / 0.000 sec

29

09:32:56

SELECT first_table.* FROM first_table INNER JOIN second_table ON first_table.id = second_table.id AND first...

1 row(s) returned

0.000 sec / 0.000 sec

30

09:32:56

SELECT first_table.name FROM first_table INNER JOIN second_table ON first_table.name = second_table.n...

4 row(s) returned

0.000 sec / 0.000 sec

31

09:32:56

SELECT first_table.* FROM first_table LEFT JOIN second_table ON first_table.id = second_table.id AND first...

1 row(s) returned

0.000 sec / 0.000 sec

32

09:32:56

SELECT first_table.name FROM first_table LEFT JOIN second_table ON first_table.name = second_table.na...

0 row(s) returned

0.000 sec / 0.000 sec

33

09:32:56

SELECT c.id, c.name, ci.address FROM class c INNER JOIN class_info ci ON c.id = ci.id LIMIT 0, 1000

3 row(s) returned

0.000 sec / 0.000 sec

34

09:32:56

SELECT c.id, c.name, CASE WHEN ci.address IS NULL THEN 'Address Missing' ELSE...

5 row(s) returned

0.000 sec / 0.000 sec